

Research Interests

VLMs, LLMs, LLM agents: evaluation, interpretability and post-training

Education

Sep 2023 – Jul 2026

University of Trento, *MSc in AI Systems*.

Thesis: *What Do Large Multimodal Models Really Know? Unlocking Knowledge Through Test-Time Computation in Open-World Image Classification*.

Advisor: *Elisa Ricci*.

- ◆ Built a 256-sample evaluation framework on the OVEN benchmark separating coverage from reliability; showed a lenient LLM judge inverts the ranking of Qwen3-VL 2B–32B, while a specificity-preserving audit restores it.
- ◆ Extended the rsbench benchmark suite to Raven's Progressive Matrices with a DeepProbLog solver, exposing a reasoning shortcut: 0.91 answer F1 while two of three concepts stayed at chance, removed with 1% concept supervision.
- ◆ Compared GPT-2 expert neurons against fMRI recordings from nine subjects using representational similarity analysis; expert neurons align with brain activity better than the dense embeddings containing them (mean $d = 0.63$).
- ◆ Improved cache-based Test-Time Adaptation with a waiting-list strategy that revisits uncertain samples, targeting non-i.i.d. streams.

2019 – 2022

University of Leeds, *BSc Computer Science with AI*.

First class. Co-led the robotics team competing at RoboCup.

Thesis: *6DoF Robot Grasping*. Advisors: *Matteo Leonetti & Mehmet Dogar*.

2017 – 2019

Marshalltown Community College, *Associate of Arts*.

Work Experience

May 2025 – Jul 2025

COVAP, Pozoblanco, Spain.

AI Engineer

Built a CrewAI agentic workflow that cut budget-document validation from 30 minutes to under 5, and a multimodal chain validating export documentation against product-tag photographs. Shipped two OpenWebUI filters and evaluated GraphRAG, docling, OCRFlux and NuExtract for the document-ingestion pipeline.

2024 – 2025

Freelance, *ML / AI Engineer*

- ◆ Benchmarked 13 open-source log parsers on accuracy, throughput, memory and tuning effort, recommending five for production; expanded a PyOD anomaly-detection benchmark from 5 to 49 algorithms with memory tracking, dimensionality reduction (MCA/UMAP) and comparative ranking.
- ◆ Customised the open-source RAGApp for optimised vector-search retrieval with an agentic workflow; authored five educational notebooks on applied AI.

Jan 2023 – Jun 2023

NEURA Robotics, Bielefeld, Germany.

Robotics Developer

Implemented ROS 2 servers for low-level control automation, explored state-of-the-art grasping techniques, and collected an RGB dataset to train a YOLO model for object detection.

Skills

Coding
Languages

Python [PyTorch, Transformers, vLLM], SLURM
Spanish, English

Competitions & Awards

2024

NASA International Space Apps, *honorable mention*; **Hub Innovazione Trentino Challenge**; **AI Scholarship**, COVAP

2023

FruitPunch AI, online

2020 – 2022

RoboCup 2021–22; **RoboCup@Home Education**, *Gold Award* 2020; **SciRoc** 2021